

Detection of Correlated Events in Log Records using Neuroevolution

José Antonio Cordón Muñoz

University of Salamanca, Department of Computer Science and Automatics

Abstract

- Cybersecurity is a continuously evolving field due to the growth of **Artificial Intelligence**, the Internet of Things (IoT), and **5G** technology, leading to more complex and advanced cyber threats.
- **Neuroevolutionary algorithms**, which optimize neural networks through an evolutionary process, have shown effectiveness in various applications but remain underutilized in cybersecurity.
- This paper proposes using the **NEAT** [1] neuroevolutionary algorithm to detect correlated events in log records from different sources, including GNU/Linux and Windows systems. The primary goal is to identify potential brute-force or ransomware attacks on a server.
- The research aims to demonstrate the effectiveness of the NEAT algorithm in enhancing defensive cybersecurity measures.

Introduction

- **Common Cyber Threats:** Brute force and ransomware attacks are common in cybersecurity [2]. Brute force attacks use dictionaries of plaintext passwords, while ransomware encrypts and makes data inaccessible.
- **Log Record Analysis:** Log records from auditing tools aid in attack detection, requiring correlation techniques to distinguish related from unrelated events.
- **Challenges in Traditional Detection:** Traditional methods require prior knowledge of event structures and specific characteristics for detection.
- **Role of AI:** The increasing complexity of cyberattacks, often using AI, necessitates more effective detection [3]. AI, particularly unsupervised learning and neuroevolutionary algorithms like NEAT, offer potential solutions.

Data Processing

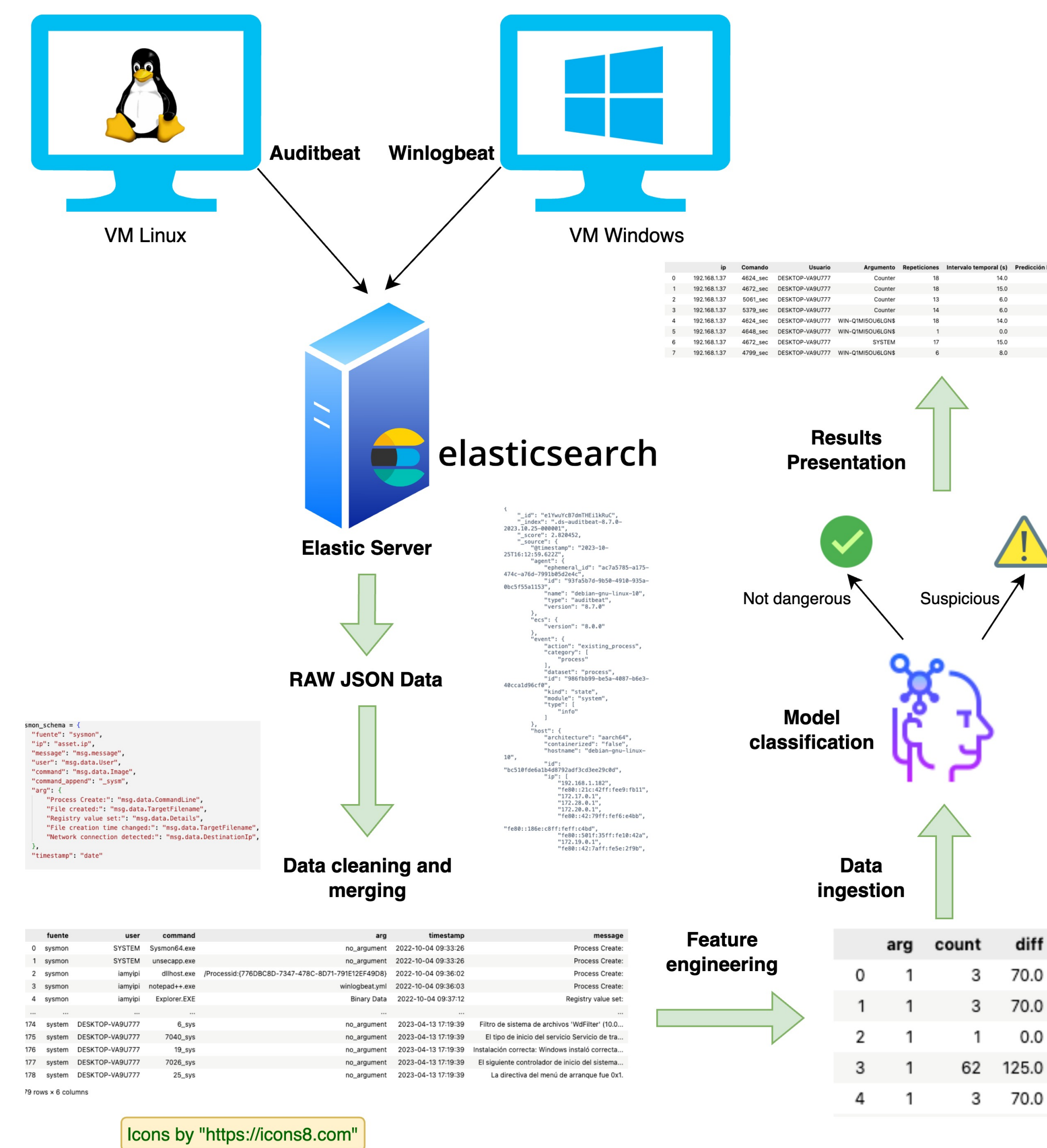


Figure 1: Data processing architecture

Data Analysis & Results

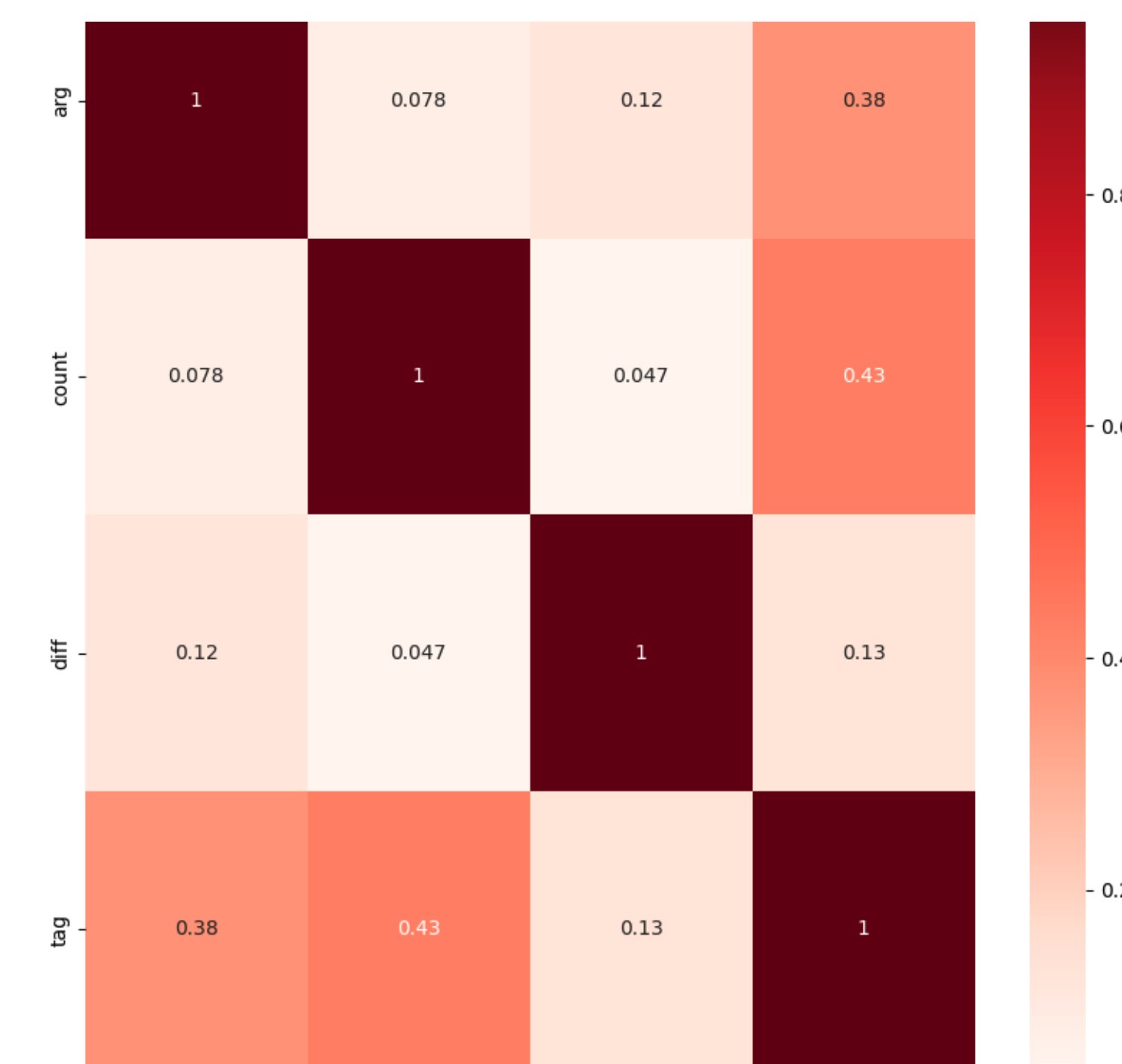


Figure 2: Correlation heat map

Source	Records (approx.)	TP / TN	FP / FN	Accuracy	Precision	TPR
Windows	1000	29 / 971	0 / 0	100%	100%	100%
Linux	5000	24 / 4976	0 / 0	100%	100%	100%

Table 1: Experiments & associated metrics

NEAT training

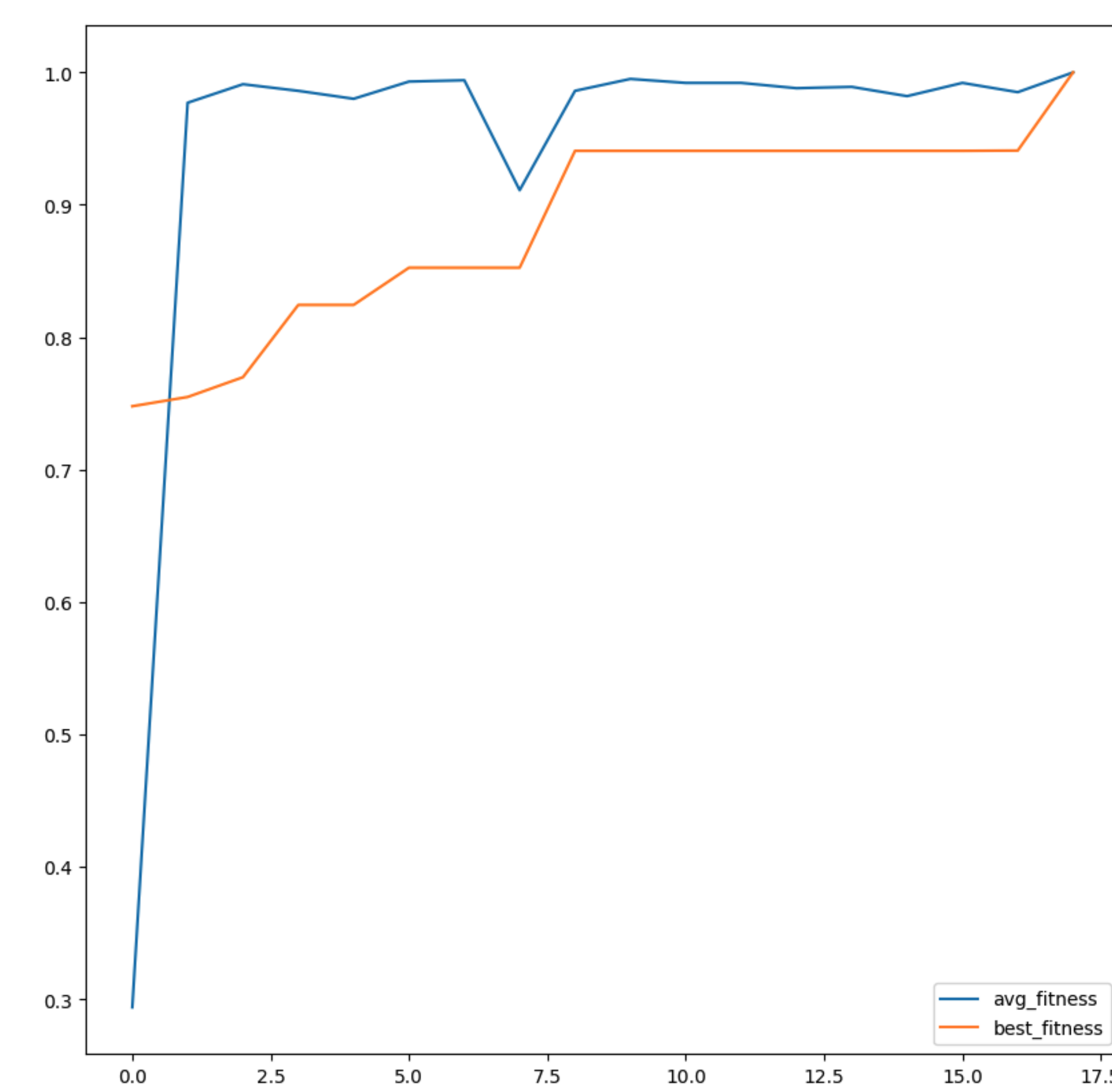


Figure 3: Fitness evolution during the training

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***** Running generation 17 *****
Population's average fitness: -3618681223557612254981338297082434661235814167665936234936185215269686721984821
Best fitness: 1.000000 - size: (2, 4) - species 3 - id 2606

Best individual in generation 17 meets fitness threshold - complexity: (2, 4)
Key: 2606
Fitness: 0.9999999999823058
Nodes:
  0 DefaultNodeGene(key=0, bias=-2.207092755631397, response=1.0, activation=sigmoid, aggregation=sum)
  200 DefaultNodeGene(key=200, bias=-3.4733884657127122, response=1.0, activation=exp, aggregation=sum)
Connections:
  DefaultConnectionGene(key=(-3, 0), weight=-0.017890261580955746, enabled=True)
  DefaultConnectionGene(key=(-2, 0), weight=-1.0987116069484126, enabled=True)
  DefaultConnectionGene(key=(-2, 200), weight=0.5428973129872413, enabled=True)
  DefaultConnectionGene(key=(-1, 0), weight=-3.4575338545209457, enabled=False)
  DefaultConnectionGene(key=(200, 0), weight=1.5249482817762817, enabled=True)
```

Figure 4: Training results

Generation 17 achieves 99% fitness, the best neural network being the one in Figure 5.

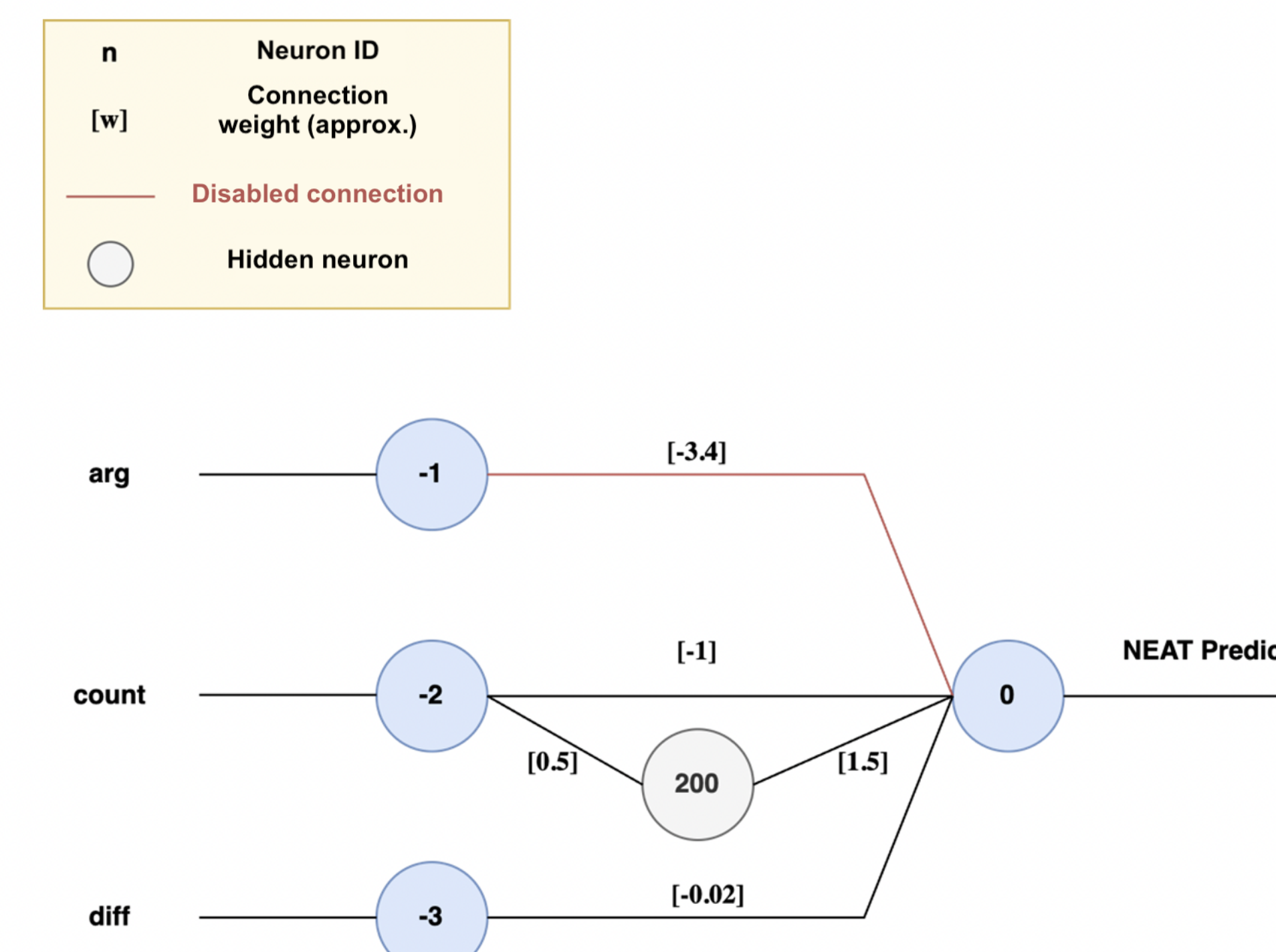


Figure 5: Optimal neural network architecture

Conclusion & Future Work

A prototype employing the NEAT algorithm for detecting correlated events has been developed, showing that:

- Neuroevolutionary algorithms demonstrate **high efficiency** in detecting correlated events in log records
 - They can provide **early attack detection** without the need for expert knowledge or complex manual configuration.
- 1 The solution's adaptability to new data sources and its capacity to detect complex correlated events involving multiple sources open opportunities for further development.
 - 2 Intrusion tests can be employed to validate the solution's effectiveness in threat detection.
 - 3 The incorporation of generative AI models to classify detected suspicious events based on their risk level has the potential to further streamline log analysis.
 - 4 Future work may explore the practical implementation of these extensions and their implications.

Related Work

- [1] Kenneth O. Stanley and Risto Miikkulainen. "Evolving Neural Networks through Augmenting Topologies". In: *Evolutionary Computation* 10 (2002), pp. 99–127.
- [2] Ahmed Fawzi Otoom, Wafa' Eleisah, and Emad E. Abdallah. "Deep Learning for Accurate Detection of Brute Force attacks on IoT Networks". In: *Procedia Computer Science* 220 (2023), pp. 291–298. ISSN: 1877-0509.
- [3] Ramanpreet Kaur, Dušan Gabrijelčič, and Tomaž Klobučar. "Artificial intelligence for cybersecurity: Literature review and future research directions". In: *Information Fusion* 97 (2023), p. 101804.